Academic Programs

UG-CSE

Program Outcomes (POs)

Engineering Graduates will be able to:

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice..

Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one?s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Educational Objectives (PEOs)

The graduates of the program will become proficient in the principles and practices of computer science, mathematics and science, enabling them to solve a wide range of computing related problems.

To enable the students with innovative applications of engineering knowledge and programming skills to spearhead the progress of society in the information age.

To mould the students into competent, successful, and practicing engineers in their career and/or in pursuing their higher studies through the spirit of innovation and entrepreneurship.

To provide exposure to cutting edge technologies, adequate training and opportunities to work individually and as teams on multidisciplinary projects with effective analytical skills.

To acquire and practice the profession with ethics, integrity and leadership qualities with due consideration to environmental issues in conformance with societal needs.

Program Specific Outcomes (PSOs)

Apply the Knowledge of Programming Languages, Networks and Databases for development of Software Applications.

Identify, Analyse, Formulate and Solve Real Time Complex Engineering Problems.

Design, Implement and Deploy a Quality Based Software System to meet the Evolving needs.

UG-CSBS

Problem analysis:

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Life-long learning:Recognize the need for,

and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

To equip students with the essential knowledge, skillsets

and attitude to be society and industry ready and make a meaningful contribution to the growth of the Indian economy.

To prepare students with fundamental concepts of Computer

Science and additionally develop an inherent capacity for liberal arts, innovative mind-set, life values and an appreciation of sustainability issues.

To prepare ?Business Engineers?, a cluster of engineering

talent tuned to the needs of Business 4.0.

To enable students to appreciate the technologies of the

future and understand the fundamental concepts of business management and develop the ?Innovation? mind-set.

To ensure students are future-ready with emerging topics

such as Cyber Security, Machine Learning, Cloud Computing, IoT, Analytics etc. as part of the curriculum

A graduate of the Computer Science and Business Systems

program will be able to:

1. Apply the fundamentals of mathematics, science and

technology abstraction to develop computational tools and applications in the areas related to algorithms, big data analytics, machine learning, and artificial intelligence and networking.

2. Leverage new age technologies to solve contemporary

challenges of varying complexity and develop ?Innovation? mind-set and entrepreneurial skills.

3. Be adept at designing business solutions using the

fundamental concepts of business management, leadership and entrepreneurial skills and appreciate life values and sustainability issues.

PG-Software Engineering

An ability to independently carry out research / investigation and development work to solve practical problems.

An ability to write and present a substantial technical report / document.

Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.

Integrate the knowledge of software engineering principles and paradigms in the design of system components and processes to meet the specific needs of the industry.

Apply the cutting-edge technologies, skills and CASE tools necessary to identify, analyze and formulate solutions to complex engineering problems with societal commitment.

Recognize the need to engage in lifelong learning that helps to explore all dimensions of software engineering practices and contemporary technologies with ethical values.

Develop technologically competent computer professionals in today?s IT centric scenario by training them in the contemporary software engineering principles and paradigms.

Provide students a deep insight into various cutting-edge technologies & tools and thereby creating diverse career opportunities

Improve analytical, logical and presentation skills of the students by applying evolving technologies of software engineering in developing practical solutions to complex problems in consonance with the legal and ethical responsibilities.

Provide the students with project engineering and management skills catering to the changing industry needs and constraints across the advancing domains of computing.

Prepare the students to take up research oriented projects, industry internships and entrepreneurship endeavours by training them to work with multi-disciplinary teams and engaging them for life-long learning in pursuit of their professional accomplishment.

Apply the knowledge of software engineering principles and paradigms in the design of system components and processes that meet the specific needs of the industry.

Use the techniques, skills and CASE tools necessary for engineering practice and coordinate the construction, deployment and maintenance of software systems.

PG-ComputerScience and Engineering

An ability to independently carry out research / investigation and development work to solve practical problems

Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program

. Apply the knowledge of engineering principles to develop software systems, products and processes thus to solve real world multifaceted problems.

Ability to design and conduct experiments, procedures and technical skills necessary for engineering exploration to solve societal problems and environmental contexts for sustainable development.

Recognize the need to engage in self-governing and life-long learning by making use of professional and ethical principles.

To provide students with sound foundations in Basic Sciences and fundamentals in Engineering Sciences.

To instil strong problem-solving skills through the courses of CSE.

To prepare students with hands on experience in implementing various software development concepts

To afford graduates with both fundamental and advanced knowledge of computer science and engineering that prepares them for excellence, leadership roles along diverse career paths, and integrate ethical behavior.

To deliver graduates to design and implement solutions for rapidly changing computing problems and information system environments and lifelong learning to adapt innovation.

Analyse software products, processes in a systematic way by applying problem solving skills and employable in product oriented Industry

Ability to take up higher studies, Research & Development and Entrepreneurships in the modern computing environment.